

SKID Modbus Communication ↔ FB_PPC_Controller Data Flow

Siemens S7-1500 · TIA Portal SCL · SMA SC 4600 UP

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1. Architecture Overview

Each SMA SC 4600 UP inverter has a dedicated SKID DB (SKID1 ... SKID10). The PPC controller maintains a 10-element **Inverter_controller** array inside **FB_PPC_Controller_DB**. Two function blocks bridge them every 100 ms OB30 scan — reading Modbus data into the SKID DB and mapping it to/from the PPC inverter array.

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SMA Inverter (Modbus TCP) → [FC04 reads, two blocks/scan]
SKIDx.db
■■ Inverter : Sunny_inverter ← FC_InputReg10_L106 (regs 10-115)
■■ Params_Inputs : Skid_Parameters_Inputs ← FC_InputReg116_L112 (regs 116-227)
■■ Params_Hold : Skid_Parameters_Hold
■■ ConnectSettings: TCON_IP_v4 (192.168.10.x : 502)

FC_PPC_SkidMapping (called 10×/scan, before FB_PPC_Controller)

FB_PPC_Controller_DB.Inverters[0..9] : Inverter_controller
■■ FC_PPC_InverterMonitor → PmaxPlant, QmaxPlant, N_Online
■■ FC_PPC_PowerDistribution → WSpt per inverter
■■ FC_PPC_ReactiveControl → VARspt / PFSpt per inverter

FC_PPC_SkidMapping writes setpoints back → SKIDx.db.Inverter
FC_Pack_Write_Regs → [FC16 writes] → SMA Inverter
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OB30 call order (critical)

- ① FC_SkidElectricStatus × 10 — DI switchgear status → FB_PPC_Controller_DB.Skids[0..9]
- ② FC_PPC_SkidMapping × 10 — reads SKID DB into Inverter_controller, writes previous scan's setpoints back
- ③ FB_PPC_Controller (InverterMonitor → RampControl → FreqResponse → PowerDistribution → ReactiveControl → FaultHandler)

2. Modbus Read Blocks — SKID DB Population

Two FCs together cover the complete SMA SC 4600 UP input register map. Both are called by ReadInverterData (FB15) once per scan per SKID.

2.1 FC InputReg10_L106_To_Inputs — SMA registers 10–115

Array offset: Reg[0] = SMA register 10. Bold rows = consumed by PPC control logic.

SMA Reg	Reg[]	Field	Scaling	Unit	Destination UDT
10..11	[0..1]	DcMs.TotWatt	S32 FIX0	W	Inps.DcMs.TotWatt
12..13	[2..3]	InvMs.DclVol.Stk1	S32 FIX1 ÷10	V	Inps.InvMs.DclVol.Stk1
14..15	[4..5]	InvMs.DclVol.Stk2	S32 FIX1 ÷10	V	Inps.InvMs.DclVol.Stk2
16..17	[6..7]	InvMs.DclVol.Stk3	S32 FIX1 ÷10	V	Inps.InvMs.DclVol.Stk3
18..19	[8..9]	InvMs.TotA.PhsA	S32 FIX0	mA	Inps.InvMs.TotA.PhsA
22..23	[12..13]	InvMs.TotA.PhsC	S32 FIX0	mA	Inps.InvMs.TotA.PhsC
24..25	[14..15]	InvMs.PF	S32 FIX4 ÷10000	—	Inps.InvMs.PF (Real)
26..27	[16..17]	InvMs.TotVA	S32 FIX0	VA	Inv.InvMs.TotVA
28..29	[18..19]	InvMs.TotW	S32 FIX0	kW	Inv.InvMs.TotW (monitoring)
30..31	[20..21]	InvMs.TotVAr	S32 FIX0	kVAr	Inv.InvMs.TotVAr (monitoring)
32..37	[22..27]	GriMs.V.PhsAB/BC/CA	S32 FIX1 ÷10	V	Inps.GriMs.V.*
38..39	[28..29]	GriMs.Hz	S32 FIX2 ÷100	Hz	Inv.GriMs.Hz (Real)
40..55	[30..45]	TmpCab.* / TmpStk.* / TmpExl / TmpTrf	S32 FIX1 ÷10	°C	Inps.Tmp*
56..69	[46..59]	DcSw1..3Stt / AcSwStt / CapacSwStt / PrchrgSwStt	S32 ENUM	—	Inps.*Stt
70..91	[60..81]	Cnt.* counters	U32/S32	s/MWh	Inps.Cnt.*
100..101	[90..91]	Rio.KeySw	S32 ENUM	—	Inps.Rio.KeySw
102..103	[92..93]	WaitGriTm	S32 FIX1 ÷10	s	Inps.WaitGriTm
104..107	[94..97]	DevInf.SerNo / GfdiSwStt	U32/S32	—	Inps.*
108..109	[98..99]	WSpt readback	S32 FIX0	kW	Inps.WSpt_Fdbk
110..111	[100..101]	VAMaxSpt	S32 FIX1 ÷10	VA	Inps.VAMaxSpt
112..113	[102..103]	VArSpt readback	S32 FIX0	kVAr	Inps.VarSpt_Fdbk
114..115	[104..105]	PFSpt readback	S32 FIX4 ÷10000	—	Inps.PFSpt_Fdbk (Real)
92..93	[82..83]	ErrLcn	U32	—	Inv.ErrLcn
94..95	[84..85]	ErrStt	S32 ENUM	—	Inv.ErrStt → Error flag
96..97	[86..87]	ErrNo	U32	—	Inv.ErrNo
98..99	[88..89]	OpStt	S32 ENUM	—	Inv.OpStt → RemReady flag

■ Green = consumed by PPC control. ■ Yellow = setpoint readback (diagnostic).

2.2 FC_InputReg116_L112_To_Inputs — SMA registers 116–227

Array offset: Reg[0] = SMA register 116.

SMA Reg	Reg[]	Field	Scaling	Unit	Destination UDT
116..123	[0..7]	TrfPro.*	S32 ENUM	—	Inps.TrfPro.*
124..135	[8..19]	TmpStk1..3.Igbt	S32 FIX1 ÷10	°C	Inps.TmpStk*.Igbt
130..135	[14..19]	TmpStk1..3.Pcb	S32 FIX1 ÷10	°C	Inps.TmpStk*.Pcb
136..137	[20..21]	Cnt.YstdWhOut	S32 FIX2 ÷100	MWh	Inps.Cnt.YstdWhOut
138..139	[22..23]	Cpu1UpTime	U32 FIX0	s	Inps.Cpu1UpTime
140..141	[24..25]	PvGnd.RisIso	S32 FIX1 ÷10	kΩ	Inps.PvGnd.RisIso
142..145	[26..29]	PresTrf / PresTrf.ErrStt	S32	—	Inps.PresTrf*
146	[30]	BfpBits	U16	—	Inps.BfpBits (Int)
147..148	[31..32]	DcMs.BfpAmp	S32 FIX1 ÷10	A	Inps.DcMs.BfpAmp
149..150	[33..34]	GriMs.Vol.PsNom	S32 FIX4 ÷10000	pu	Inps.GriMs.Vol.PsNom
151..158	[35..42]	Cnt.TotDcWhOut / TotAcWhIn / DcWhOut / AcWhIn	S32 FIX2 ÷100	MWh	Inps.Cnt.*
159..160	[43..44]	DclVolSpt	S32 FIX1 ÷10	V	Inps.DclVolSpt
163..164	[47..48]	VolNomSpt	S32 FIX4 ÷10000	pu	Inps.VolNomSpt
167	[51]	AuxCtl.LifeSign	U16	—	Inps.AuxCtl.LifeSign (Int)
168..169	[52..53]	InvPwrVolTyp	S32 ENUM	—	Inps.InvPwrVolTyp
170..171	[54..55]	HzNomSpt	S32 FIX2 ÷100	Hz	Inps.HzNomSpt
172..173	[56..57]	WAval	S32 FIX4 ÷10000	pu	Inv.WAval (Real 0.0–1.0)
174..175	[58..59]	VArAval	S32 FIX4 ÷10000	pu	Inv.VArAval (Real 0.0–1.0)
176..177	[60..61]	DrtStt	S32 ENUM	—	Inps.DrtStt
178..179	[62..63]	PwrOffReas	S32 ENUM	—	Inv.PwrOffReas
180..181	[64..65]	DiagRmgTm	U32 FIX0	s	Inps.DiagRmgTm
182	[66]	Rio.Din.FloatCtl.Wa	U16	—	Inps.Rio.Din.FloatCtl.Wa
183	[67]	Rio.Din.FloatCtl.Err	U16	—	Inps.Rio.Din.FloatCtl.Err
184..185	[68..69]	WRtg	S32 FIX0	kW	Inps.WRtg (commissioning read)
186	[70]	Gfdi.AmpPrc	S16 FIX2 ÷100	A	Inps.Gfdi.AmpPrc
187	[71]	Gfdi.AmpErr	S16 FIX2 ÷100	A	Inps.Gfdi.AmpErr
188..227	[72..111]	ActErrNo1..10 / ActErrLcn1..10	U32	—	Inps.ActErrNo/Lcn 1..10

■ Green = consumed by PPC control logic.

3. FC_PPC_SkidMapping — The Bridge

Called once per SKID in OB30 (10 calls total), **before** FB_PPC_Controller. Performs read and write directions in a single call. Skips setpoint writes when CommError = TRUE.

3.1 READ direction: SKID DB → Inverter_controller

Source (Sunny_inverter / Params_Inputs)	Conversion	Destination (Inverter_controller)	Note
SkidInverter.OpStt (DInt ENUM)	= 3526 OR 3527 OR 3529 OR 3530	Inverter.RemReady (Bool)	3526=GridFeed, 3527=FRT, 3529=QonDemand, 3530=RampDown
SkidInverter.ErrStt (DInt ENUM)	≠ 307	Inverter.Error (Bool)	307 = Ok (no fault)
SkidInverter.InvMs.TotW (DInt, kW)	direct copy	Inverter.Wactive (DInt, kW)	monitoring only — not used in control
SkidInverter.InvMs.TotVAr (DInt, kVAr)	direct copy	Inverter.Qactive (DInt, kVAr)	monitoring only
SkidInverter.WAval (Real, pu 0.0–1.0)	× 10000 → REAL_TO_DINT	Inverter.WAval (DInt, FIX4)	drives PmaxPlant and proportional distribution
SkidInverter.VArAval (Real, pu 0.0–1.0)	× 10000 → REAL_TO_DINT	Inverter.VArAval (DInt, FIX4)	drives QmaxPlant
SkidInverter.PwrOffReas (DInt ENUM)	direct copy	Inverter.PwrOffReas (DInt)	≠ 0 → not Available (online criteria)
SkidInputs.DrtStt (DInt ENUM)	direct copy	Inverter.DrtStt (DInt)	derating state — diagnostic
CommError (Bool)	direct copy	Inverter.CommError (Bool)	online criteria

■ Green = primary dispatch fields.

3.2 WRITE direction: Inverter_controller → SKID DB (skipped if CommError)

Source (Inverter_controller)	Conversion	Destination (Sunny_inverter)	SMA Reg	Note
Inverter.OperMode (DInt)	direct	SkidInverter.RemRdy	1st	SMA requires RemRdy before InvOpMod
Inverter.OperMode (DInt)	direct	SkidInverter.InvOpMod	2nd	308=Run, 303=Stop
Inverter.WMode (DInt)	direct	SkidInverter.GriMng.WMod	0 grp	1079=WCtlCom, 303=Off
Inverter.WSpt (DInt, kW)	direct	SkidInverter.WSpt	108	active power setpoint
Inverter.VArMode (DInt)	direct	SkidInverter.GriMng.VArMod	0 grp	1072=VArCtlCom, 1075=PFCtlCom
Inverter.VArSpt (DInt, kVAr)	direct	SkidInverter.VArSpt	112	reactive power setpoint
Inverter.PFSpt (DInt, FIX4)	÷ 10000 → Real	SkidInverter.PFSpt (Real, cos ϕ)	114	FC_Pack_Write_Regs packs back to FIX4 for FC16
Inverter.ErrClr (DInt)	direct	SkidInverter.ErrClr	8	one-shot: 26 for one scan, then 0

4. WAval / VArAval Scaling Chain (End-to-End)

This is the most critical scaling path — it drives active power dispatch, proportional distribution, and the FreqResponse dynamic band.

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SMA Modbus register 172 (S32 FIX4)
Raw example: 8500 → 85 % of rated power available

FC_InputReg116_L112_To_Inputs:
Inv.WAval := DINT_TO_REAL(di32) / 10000.0
↓
SKID1.Inverter.WAval = 0.85 (Real, per-unit 0.0-1.0)

FC_PPC_SkidMapping:
Inverter.WAval := REAL_TO_DINT(SkidInverter.WAval × 10000.0)
↓
FB_PPC_Controller_DB.Inverters[0].WAval = 8500 (DInt, FIX4 pu×10000)

FC_PPC_InverterMonitor:
PmaxPlant += DINT_TO_REAL(WAval) / 10000.0 × WRtg_kw
= 8500 / 10000.0 × 4600 kW = 3910 kW (this inverter's contribution)
↓
PmaxPlant (Real, kW) = sum across all Available inverters → up to 46 000 kW

FB_PPC_FreqResponse:
Pmax_active = Pmax_disp - dP_at_200mHz_OF (Pmax_disp = PmaxPlant)
↓
FC_PPC_PowerDistribution:
share_i = WAval[i] / ΣWAval[j] (raw DInt ratio – units cancel)
WSpt[i] = P_final_kw × share_i → per-inverter kW setpoint
maxWSpt = WAval[i] / 10000 × WRtg_kw → per-inverter clamp
↓
Inverter_controller.WSpt[0..9] (DInt, kW) → written via FC_PPC_SkidMapping
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Note: In the proportional split, the ÷10000 and ×WRtg_kw factors cancel across numerator and denominator, so raw DInt FIX4 values are used directly for the ratio — no conversion at that stage. Conversion is only needed for absolute kW/kVAr values (PmaxPlant, QmaxPlant, per-inverter WSpt clamp).

5. InverterMonitor Online Criteria

FC_PPC_InverterMonitor computes Inverters[i].Available once per scan. All six conditions must be TRUE. Result is cached into Available — PowerDistribution, ReactiveControl, and FaultHandler read this, never re-deriving it.

Condition	Source field	Set to FALSE when
Enabled	DB39 / HMI write	Operator has disabled this inverter slot
RemReady	Inverter.RemReady (from OpStt)	OpStt ∉ {3526, 3527, 3529, 3530}
NOT Error	Inverter.Error (from ErrStt)	ErrStt ≠ 307
NOT CommError	Inverter.CommError (from MB_CLIENT)	Modbus TCP timeout or error
PwrOffReas = 0	Inverter.PwrOffReas (from OpStt group)	Inverter disconnected abnormally
ELECTRIC_OK	FB_PPC_Controller_DB.Skids[i]	Breaker / separator / earthing not OK (DI status)

6. Scaling Conventions Summary

Signal	SMA Modbus	Sunny_inverter (SKID DB)	Inverter_controller (PPC)	PPC computation
WAval	S32 FIX4 reg 172	Real pu ($\div 10000$)	DInt FIX4 ($\times 10000$)	$\div 10000 \times \text{WRtg_kW} \rightarrow \text{kW}$
VArAval	S32 FIX4 reg 174	Real pu ($\div 10000$)	DInt FIX4 ($\times 10000$)	$\div 10000 \times \text{WRtg_kW} \rightarrow \text{kVAr}$
InvMs.TotW	S32 FIX0 reg 28, kW	DInt kW (direct)	DInt kW (direct)	monitoring only
InvMs.TotVAr	S32 FIX0 reg 30, kVAr	DInt kVAr (direct)	DInt kVAr (direct)	monitoring only
GriMs.Hz	S32 FIX2 reg 38	Real Hz ($\div 100$)	—	read by PPC via Sunny_inverter.GriMs.Hz
WSpt	S32 FIX0 reg 108, kW	DInt kW (PPC writes)	DInt kW (direct)	readback $\rightarrow$ Inps.WSpt_Fdbk
VArSpt	S32 FIX0 reg 112, kVAr	DInt kVAr (PPC writes)	DInt kVAr (direct)	readback $\rightarrow$ Inps.VarSpt_Fdbk
PFSpt	S32 FIX4 reg 114	Real $\cos \phi$ (FC_Pack $\times 10000$)	DInt FIX4 e.g. 9500	readback $\div 10000 \rightarrow$ Inps.PFSpt_Fdbk
WRtg	S32 FIX0 reg 184, kW	DInt kW (Inps.WRtg)	—	DB39.WRtg_kW = 4600 (commissioning)
OpStt	S32 ENUM reg 98	DInt ENUM (direct)	$\rightarrow$ RemReady (Bool)	3526/3527/3529/3530 $\rightarrow$ TRUE
ErrStt	S32 ENUM reg 94	DInt ENUM (direct)	$\rightarrow$ Error (Bool)	$\neq 307 \rightarrow$ TRUE
PwrOffReas	S32 ENUM reg 178	DInt ENUM (direct)	DInt (direct)	$\neq 0 \rightarrow$ not Available
DrtStt	S32 ENUM reg 176	DInt ENUM (Params_Inputs)	DInt (direct)	derating diagnostic

7. Key Constants — DB39 (PPC_Controller)

Parameter	Value	Source / Note
WRtg_kW	4600.0 kW	Read from SMA reg 184 at commissioning; stored in DB39
Pn_MW	46.0 MW	$10 \times \text{WRtg_kW} \div 1000$ (plant nameplate)
Droop_OF_pct	8.0 %	ANRE Ord.51/2019 Art. 114-115 (over-frequency)
Droop_UF_pct	10.0 %	ANRE Ord.51/2019 Art. 118-120 (under-frequency)